ASPREDICTED

Wharton CREDIBILITY LAB
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Viscereality – Symmetric Breathing Space (#262545)

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Pre-registered on:
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1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

The primary aim of this study is to evaluate the efficacy of the Viscereality VR system (Fejer et al., 2025) in inducing alterations in subjective state of consciousness, self-related experience (including the sense of self and self-boundaries), and adherence to box-breathing instructions. The VR system implements a 3D bioresponsive particle environment that contracts and expands in real time according to changes in lung volume measured via the VR controller. The particle field's visual properties (e.g., size, color, texture, animation cycle) are parametrically varied to generate continuous, dynamic visual patterns. By systematically manipulating the algorithm governing the phase dynamics of the particle animations via coupled oscillators, we vary the degree of visual symmetry presented to participants. These parameters are used to construct two distinct VR scenarios that differ in their degree of dynamic visual (a)symmetry.

Our primary aim is to provide a first validation of the previously published Viscereality VR system and assess its ability to induce alterations in participants' state of consciousness during breathwork, as well as its potential to support adherence to breathing instructions.

A secondary aim of the study is to examine whether dynamic visual symmetry modulates the subjective experience, particularly its affective valence. We hypothesize that higher levels of visual symmetry will be associated with more positively valenced subjective ratings. We further hypothesize that visual symmetry may enhance participant engagement, improve adherence to breathing instructions, and contribute to stronger alterations in self-related experience, including a greater reduction in the sense of self and perceived self-boundaries.

Fejer G, Holzapfel T, Gómez-Emilsson A, Hirvonen T, Jermaks R, Blum J, Lobser D, Lalidis Mateo A, Glowacki D, Gaebler M, Lenggenhager B. Viscereality: A Bio-responsive VR System for Breath-Based Interactions and Coupled Oscillator Dynamics to Augment Altered States of Consciousness. Mensch und Computer 2025-Workshopband 2025 (pp. 10-18420). Gesellschaft für Informatik eV.

3) Describe the key dependent variable(s) specifying how they will be measured.

The study includes five families of dependent variables.

(1) Altered states of consciousness will be measured using the 11-Dimension Altered States of Consciousness Rating Scale (11D-ASC; Studerus et al., 2010), consisting of 42 items rated on 0–100 sliders, yielding one score per dimension and condition.

(2) Temporal dynamics of subjective experience will be assessed using temporal experience tracers adapted for VR. Participants continuously rate the intensity of each ASC dimension throughout each breathing block, producing time-series data. Comparable methods have tracked experiential dynamics during breathwork (Lewis-Healy et al., 2024) and meditation (Jachs et al., 2022).

(3) Self-related experience will be measured using three pictographic instruments: the Perceived Body Boundaries Scale (Dambrun, 2016) indexing boundary permeability, the Spatial Frame of Reference Continuum (Hanley et al., 2020) measuring spatial extension of self, and the Small Self Scale (Vidal et al., 2024) assessing diminished self-salience.

(4) Adherence to breathing instructions will be quantified as the sum of absolute deviations from the instructed 4-second duration across all four phases of each breath cycle, averaged across valid cycles per condition.

(5) Blinding will be assessed via an ad hoc 12-item manipulation awareness questionnaire and a forced-choice guess task with confidence rating, quantified using Bang's Blinding Index (Bang et al., 2004) and certainty ratings.

4) How many and which conditions will participants be assigned to?

The study uses a within-subjects design with three conditions, completed by all participants in counterbalanced order.

High-symmetry VR condition:

Participants perform 4–4–4–4 box breathing within a bioresponsive particle environment whose dynamic visual symmetry is generated via a coupling kernel applied to the system's oscillator dynamics. When the kernel is configured for strong harmonic coupling, the oscillators synchronize and drive coherent changes across visual features of the particle field, including size modulations, color shifts, motion trajectories, and other dynamic attributes. This produces a highly symmetrical and unified visual structure (Fejer et al., 2025).

No-symmetry VR condition:

Participants perform the same breathing task within an otherwise identical particle environment, but the coupling kernel is modified to reduce symmetry. By weakening harmonic coupling and introducing destabilizing or weakly structured interactions among oscillators, the system prevents coherent synchronization of particle features. As a result, changes in particle size, color, motion, and other attributes remain desynchronized and visually non-symmetrical (Fejer et al., 2025).

Blackscreen control condition:

Participants wear the VR headset but view a completely black display while performing the same 4–4–4–4 box-breathing task guided by standardized audio instructions.

Breathing instructions, task duration, and respiratory signal acquisition via the VR controllers remain identical across all three conditions.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

Altered States of Consciousness: For each of the eleven ASC dimensions, we will conduct one-way RM-ANOVA with condition (symmetric VR, asymmetric VR, blackscreen control) as the within-subjects factor and presentation order as a covariate. Two planned contrasts will be applied: pooled VR conditions versus blackscreen control, and symmetric versus asymmetric VR. Bayesian RM-ANOVAs will be conducted in parallel using JASP. For dimensions with specific directional predictions (Experience of Unity, Blissful State, Audio-Visual Synesthesia), we will compute Bayes factors for the predicted ordering (symmetric VR > asymmetric VR > blackscreen).

Temporal Experience Tracers: From each participant's continuous intensity trace, we extract peak intensity and aggregate intensity (area under the curve). We will compute correlations between these metrics and corresponding 11D-ASC scores to assess convergent validity. Both metrics will also be analyzed using the same RM-ANOVA approach as the primary analyses.

Self-Related Experience: The Perceived Body Boundaries Scale, Spatial Frame of Reference Continuum, and Small Self Scale will each be analyzed using one-way RM-ANOVA with condition and presentation order as covariate, applying the same contrasts. Bayesian RM-ANOVAs conducted in parallel.

Breathing Adherence: Composite adherence error per cycle is computed as the sum of absolute deviations from 4-second targets across all four phases. Valid cycles require all phases detectable and within 2–8 seconds. Mean composite error per condition analyzed via RM-ANOVA with order covariate.

Blinding Integrity: A 12-item manipulation awareness questionnaire assesses detection of true differences, with sham items indexing response bias. Following unblinding, participants guess which scenario was more symmetric with confidence rating. Bang's Blinding Index (Bang et al., 2004) quantifies blinding success, which we compute with and without confidence ratings.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Participants will be excluded according to the following a priori criteria:

Medical and safety exclusions apply to participants with self-reported cardiovascular conditions (e.g., arrhythmia, hypertension requiring medication), self-reported respiratory conditions (e.g., asthma, COPD, chronic breathing difficulties), current pregnancy, or history of severe motion sickness or vestibular disorders.

Withdrawal during the study will result in exclusion for participants who discontinue participation for any reason and do not complete all three experimental blocks.

Incomplete data exclusions apply to participants who do not complete all three experimental blocks (symmetric VR, asymmetric VR, blackscreen control) or who have missing or unusable respiratory data (e.g., controller tracking loss, signal dropout).

Non-adherence to the box-breathing protocol will be evaluated based on two criteria. First, valid cycles are defined as complete four-phase sequences (inhale-hold-exhale-hold) in which all phases are detectable from the respiratory signal and fall within a plausible duration range (2–8 seconds per phase). Participants will be excluded if fewer than 70% of their breathing cycles across all blocks meet these validity criteria. Second, for participants meeting the validity threshold, a mean adherence error will be computed across all valid cycles. Participants will be excluded if their overall adherence error is more than 2 standard deviations above the sample mean, calculated after excluding participants removed for other reasons. These criteria ensure that retained participants both engaged with the box-breathing structure and followed the 4–4–4–4 timing with reasonable accuracy.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We aim to obtain 36 complete datasets, with 6 participants in each of the 6 counterbalancing orders. Participants who do not complete all blocks or whose data are unusable (including technical failures or non-adherence as defined above) will be replaced until all orders are fully filled. We view this sample size as adequate for the within-subjects design and the expected medium effect sizes.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

At the time of preregistration, data from 19 participants had already been collected. These data were obtained during a pilot phase designed to verify that the Meta Quest device was logging all signals correctly and to troubleshoot technical issues such as GPU overheating during continuous full-day experimental use. The pilot was conducted through a student pool, and due to unexpectedly high interest, 19 students volunteered to participate. The data have not been inspected beyond confirming successful data logging. Because finalizing the preregistration required additional time to obtain approval from all main authors, approximately half of the planned sample was collected before preregistration approval was completed.

BUNDLE

This pre-registration is part of a bundle which includes:
#262,919 - https://aspredicted.org/za3zq6.pdf - Title: 'Visceral Breathing Space'